Volume estimation of food from 2D images

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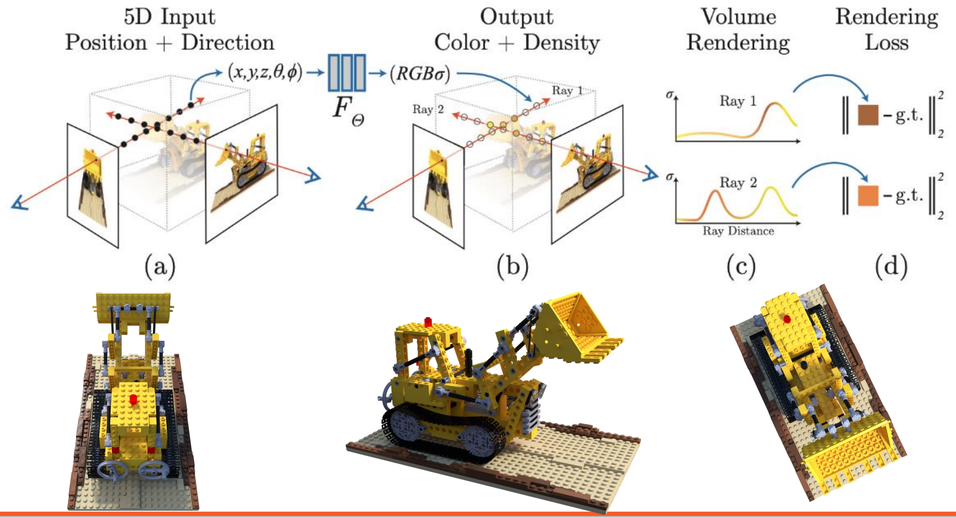
2020

Objective:

- Explore Various methods of volume estimation from 2D images of food
- Compare results obtained

Proposed Methodologies:

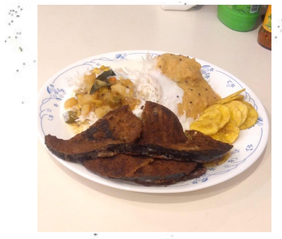
- Computer Vision Techniques
  - Birds Eye View
  - Frame Transform Methods
  - Visual SFM
- Deep Learning Techniques
  - Fully Connected Network (Neural Radiance Fields)
  - GANs with CNN generators as generator



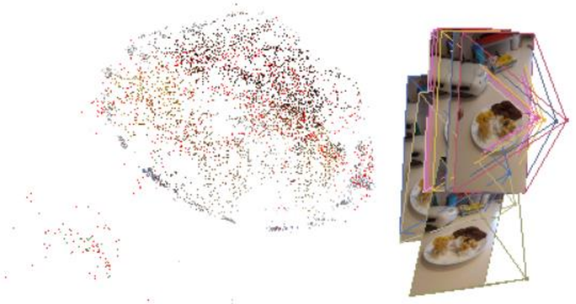
Results:

Computer-vision Technique

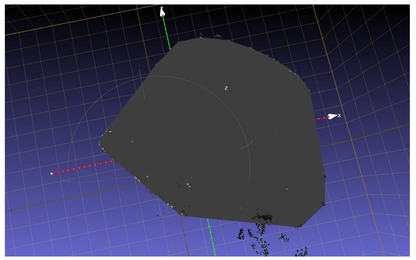
- **Visual SFM (structure from motion)**







Uses SIFT , feature detection, feature matching and multicore bundle adjustments



- Convex hull rendered in Mesh Lab for volume estimation

Deep Learning Technique

- **Fully connected network**
  - Works well with synthetic data
  - Takes a long time for rendering real images

Future scope :

- Use Neural Radiance field Network with real images of good to obtain 3D models
- Train a suitable GAN model for 3D reconstruction of irregular objects

Conclusion

- At present with all the restrictions, computer vision techniques perform better than the deep learning techniques